

# Cambridge International AS & A Level

---

**MATHEMATICS****9709/11**

Paper 1 Pure Mathematics 1

**May/June 2025****MARK SCHEME**

Maximum Mark: 75

---

Published

---

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2025 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

---

This document consists of **24** printed pages.

**PUBLISHED****Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

**GENERIC MARKING PRINCIPLE 1:**

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

**GENERIC MARKING PRINCIPLE 2:**

Marks awarded are always **whole marks** (not half marks, or other fractions).

**GENERIC MARKING PRINCIPLE 3:**

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

**GENERIC MARKING PRINCIPLE 4:**

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

**GENERIC MARKING PRINCIPLE 5:**

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

**GENERIC MARKING PRINCIPLE 6:**

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

**Mathematics-Specific Marking Principles**

- 1 Unless a particular method has been specified in the question, full marks may be awarded for any correct method. However, if a calculation is required then no marks will be awarded for a scale drawing.
- 2 Unless specified in the question, non-integer answers may be given as fractions, decimals or in standard form. Ignore superfluous zeros, provided that the degree of accuracy is not affected.
- 3 Allow alternative conventions for notation if used consistently throughout the paper, e.g. commas being used as decimal points.
- 4 Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored (isw).
- 5 Where a candidate has misread a number or sign in the question and used that value consistently throughout, provided that number does not alter the difficulty or the method required, award all marks earned and deduct just 1 A or B mark for the misread.
- 6 Recovery within working is allowed, e.g. a notation error in the working where the following line of working makes the candidate's intent clear.

**Annotations guidance for centres**

Examiners use a system of annotations as a shorthand for communicating their marking decisions to one another. Examiners are trained during the standardisation process on how and when to use annotations. The purpose of annotations is to inform the standardisation and monitoring processes and guide the supervising examiners when they are checking the work of examiners within their team. The meaning of annotations and how they are used is specific to each component and is understood by all examiners who mark the component.

We publish annotations in our mark schemes to help centres understand the annotations they may see on copies of scripts. Note that there may not be a direct correlation between the number of annotations on a script and the mark awarded. Similarly, the use of an annotation may not be an indication of the quality of the response.

The annotations listed below were available to examiners marking this component in this series.

**Annotations**

| <b>Annotation</b>                                                                   | <b>Meaning</b>                         |
|-------------------------------------------------------------------------------------|----------------------------------------|
|    | More information required              |
|    | Accuracy mark awarded zero             |
|    | Accuracy mark awarded one              |
|    | Independent accuracy mark awarded zero |
|   | Independent accuracy mark awarded one  |
|  | Independent accuracy mark awarded two  |
|  | Benefit of the doubt                   |
|  | Blank Page                             |
|  | Incorrect                              |
| Dep                                                                                 | Used to indicate DM0 or DM1            |

| Annotation                                                                          | Meaning                                                                                                                                    |
|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| DM1                                                                                 | Dependent on the previous M1 mark(s)                                                                                                       |
|    | Follow through                                                                                                                             |
|    | Indicate working that is right or wrong                                                                                                    |
| Highlighter                                                                         | Highlight a key point in the working                                                                                                       |
|    | Ignore subsequent work                                                                                                                     |
|    | Judgement                                                                                                                                  |
|    | Judgement                                                                                                                                  |
|    | Method mark awarded zero                                                                                                                   |
|    | Method mark awarded one                                                                                                                    |
|    | Method mark awarded two                                                                                                                    |
|    | Misread                                                                                                                                    |
|    | Omission or Other solution                                                                                                                 |
| Off-page comment                                                                    | Allows comments to be entered at the bottom of the RM marking window and then displayed when the associated question item is navigated to. |
| On-page comment                                                                     | Allows comments to be entered in speech bubbles on the candidate response.                                                                 |
|  | Judgment made by the PE                                                                                                                    |
|  | Premature approximation                                                                                                                    |
|  | Special case                                                                                                                               |
|  | Indicates that work/page has been seen                                                                                                     |

| Annotation                                                                        | Meaning                                |
|-----------------------------------------------------------------------------------|----------------------------------------|
|  | Error in number of significant figures |
|  | Correct                                |
|  | Transcription error                    |
|  | Correct answer from incorrect working  |

**PUBLISHED****Mark Scheme Notes**

The following notes are intended to aid interpretation of mark schemes in general, but individual mark schemes may include marks awarded for specific reasons outside the scope of these notes.

**Types of mark**

- M** Method mark, awarded for a valid method applied to the problem. Method marks are not lost for numerical errors, algebraic slips or errors in units. However, it is not usually sufficient for a candidate just to indicate an intention of using some method or just to quote a formula; the formula or idea must be applied to the specific problem in hand, e.g. by substituting the relevant quantities into the formula. Correct application of a formula without the formula being quoted obviously earns the M mark and in some cases an M mark can be implied from a correct answer.
- A** Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. Accuracy marks cannot be given unless the associated method mark is earned (or implied).
- B** Mark for a correct result or statement independent of method marks.
- DM or DB** When a part of a question has two or more ‘method’ steps, the M marks are generally independent unless the scheme specifically says otherwise; and similarly, when there are several B marks allocated. The notation DM or DB is used to indicate that a particular M or B mark is dependent on an earlier M or B (asterisked) mark in the scheme. When two or more steps are run together by the candidate, the earlier marks are implied and full credit is given.
- FT** Implies that the A or B mark indicated is allowed for work correctly following on from previously incorrect results. Otherwise, A or B marks are given for correct work only.
- A or B marks are given for correct work only (not for results obtained from incorrect working) unless follow through is allowed (see abbreviation FT above).
  - For a numerical answer, allow the A or B mark if the answer is correct to 3 significant figures or would be correct to 3 significant figures if rounded (1 decimal place for angles in degrees).
  - The total number of marks available for each question is shown at the bottom of the Marks column.
  - Wrong or missing units in an answer should not result in loss of marks unless the guidance indicates otherwise.
  - Square brackets [ ] around text or numbers show extra information not needed for the mark to be awarded.

**Abbreviations**

|        |                                                                                                                                                                                       |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AEF/OE | Any Equivalent Form (of answer is equally acceptable) / Or Equivalent                                                                                                                 |
| AG     | Answer Given on the question paper (so extra checking is needed to ensure that the detailed working leading to the result is valid)                                                   |
| CAO    | Correct Answer Only (emphasising that no ‘follow through’ from a previous error is allowed)                                                                                           |
| CWO    | Correct Working Only                                                                                                                                                                  |
| ISW    | Ignore Subsequent Working                                                                                                                                                             |
| SOI    | Seen Or Implied                                                                                                                                                                       |
| SC     | Special Case (detailing the mark to be given for a specific wrong solution, or a case where some standard marking practice is to be varied in the light of a particular circumstance) |
| WWW    | Without Wrong Working                                                                                                                                                                 |
| AWRT   | Answer Which Rounds To                                                                                                                                                                |



| Question | Answer                                                                                        | Marks        | Guidance                                                                                                                                                                                                                                                           |
|----------|-----------------------------------------------------------------------------------------------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        | $6\sin^2 \theta - \sin \theta - 2 = 0 \Rightarrow [(2\sin \theta + 1)(3\sin \theta - 2) = 0]$ | <b>M1</b>    | For expressing as a 3-term quadratic.<br>Terms need not all be on the same side.                                                                                                                                                                                   |
|          | $\sin \theta = -\frac{1}{2}$ or $\sin \theta = \frac{2}{3}$                                   | <b>A1</b>    | For both.<br>Allow AWRT 0.667.                                                                                                                                                                                                                                     |
|          | $\theta = -150^\circ, -30^\circ, 41.8^\circ, 138.2^\circ$                                     | <b>A1 A1</b> | AWRT<br>A1 for any correct angle from a correct value of $\sin \theta$ , A1 for all 4 and no others in the interval $-180^\circ < \theta < 180^\circ$ .<br><b>SC B1</b> for $\frac{-5\pi}{6}, \frac{-\pi}{6}, 0.730^\circ, 2.41^\circ$ if use of radians is clear. |
|          |                                                                                               | <b>4</b>     |                                                                                                                                                                                                                                                                    |

| Question | Answer                                                                                 | Marks        | Guidance                                                                                                                                                |
|----------|----------------------------------------------------------------------------------------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2(a)     | $[\text{Gradient of tangent}] = 4(2 \times 4 - 5)^3 - 9 \times 4^{\frac{1}{2}} [= 90]$ | <b>M1</b>    | Substitute $x = 4$ into $\frac{dy}{dx}$ .<br>$\frac{-11}{2} = 4(2 \times 4 - 5)^3 - 9 \times 4^{\frac{1}{2}}$ is M0 unless they reach $-\frac{1}{90}$ . |
|          | $[\text{Gradient of normal}] = -\frac{1}{90}$                                          | <b>A1</b>    | AWRT $-0.0111$ .                                                                                                                                        |
|          |                                                                                        | <b>2</b>     |                                                                                                                                                         |
| 2(b)     | $[y] = \left\{ \frac{1}{2}(2x - 5)^4 \right\} \left\{ -6x^{\frac{3}{2}} \right\} [+c]$ | <b>B1 B1</b> | Accept unsimplified.                                                                                                                                    |
|          | $-\frac{11}{2} = \frac{1}{2} \times (2 \times 4 - 5)^4 - 6 \times 4^{\frac{3}{2}} + c$ | <b>M1</b>    | Sub $x = 4, y = -\frac{11}{2}$ into an integrated expression and attempt to find $c$ .                                                                  |
|          | $y = \frac{1}{2}(2x - 5)^4 - 6x^{\frac{3}{2}} + 2$                                     | <b>A1</b>    | Condone $c = 2$ as final answer if ‘ $y = \dots$ ’ seen previously.<br>Fractions must be simplified.<br>Accept $f(x)$ in place of $y$ .                 |
|          |                                                                                        | <b>4</b>     |                                                                                                                                                         |

| Question | Answer                                                                                                      | Marks     | Guidance                                                                                                                                                     |
|----------|-------------------------------------------------------------------------------------------------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3(a)     | $ar^2 = 18, a + ar + ar^2 = 26$ or $a + ar + 18 = 26$                                                       | <b>B1</b> | For first and second, or first and third equations.                                                                                                          |
|          | $\frac{18}{r^2} + \frac{18}{r} + 18 = 26 \Rightarrow 8r^2 - 18r - 18 = 0 \Rightarrow [(4r + 3)(r - 3) = 0]$ | <b>M1</b> | For expressing as a 3-term quadratic from two expressions with sign errors only. Terms need not all be on one side.                                          |
|          | $r = -\frac{3}{4}$ [or $r = 3$ ]                                                                            | <b>A1</b> | CAO<br>Allow $-0.75$ .                                                                                                                                       |
|          | $a = 32$                                                                                                    | <b>A1</b> | CAO<br>Ignore $a = 2$ .<br>May not find $a$ , but instead use expression for $a$ in terms of $r$ (e.g. $a = \frac{18}{r^2}$ ) and then find $a$ in part (b). |
|          | Tenth term = $-2.40$                                                                                        | <b>A1</b> | AWRT. Ignore other values.                                                                                                                                   |
|          | <b>Alternative Method for first three marks in Question 3(a)</b>                                            |           |                                                                                                                                                              |
|          | $ar^2 = 18, 26 = \frac{a(1-r^3)}{1-r}$                                                                      | <b>B1</b> |                                                                                                                                                              |
|          | $26 = \frac{\frac{18}{r^2}(1-r^3)}{1-r} \Rightarrow 8r^3 - 26r^2 + 18 = 0$                                  | <b>M1</b> | Expressing as a 3-term cubic from two expressions with sign errors only. Terms need not all be on one side                                                   |
|          | $r = -\frac{3}{4}$ [or $r = 3, 1$ ]                                                                         | <b>A1</b> | CAO<br>Accept $-0.75$ .                                                                                                                                      |
|          |                                                                                                             | <b>5</b>  | Note: <b>SC B1 B1 B1</b> is possible following B1 M0 if no method is shown for finding $r = -\frac{3}{4}$ [or $r = 3$ ].                                     |

| Question | Answer                                                  | Marks        | Guidance                                                                        |
|----------|---------------------------------------------------------|--------------|---------------------------------------------------------------------------------|
| 3(b)     | $S_{\infty} = \frac{32}{1 - \left(-\frac{3}{4}\right)}$ | <b>M1</b>    | FT on values of $a$ and $r$ , provided $-1 < r < 1$ .                           |
|          | $\frac{128}{7}$                                         | <b>A1 FT</b> | FT on values of $a$ and $r$ , provided $-1 < r < 0$ .<br>Condone extra answers. |
|          |                                                         | <b>2</b>     |                                                                                 |

| Question | Answer                                                                                                                                                          | Marks        | Guidance                                                                                                                                                                                                                                                                                                                           |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4        | Attempt to integrate $(x-16) - \left(5x^{\frac{3}{2}} - 20x\right) \left[ -5x^{\frac{3}{2}} + 21x - 16 \right]$                                                 | <b>*M1</b>   | Attempt to integrate both terms and subtract areas. Accept subtraction either way round.                                                                                                                                                                                                                                           |
|          | $\left\{ \frac{1}{2}x^2 - 16x \right\} [-] \left\{ \left( 2x^{\frac{5}{2}} - 10x^2 \right) \right\} \left[ \right] = -2x^{\frac{5}{2}} + \frac{21}{2}x^2 - 16x$ | <b>B1 B1</b> | B1 for each integral.<br>B2,1, 0 for $-2x^{\frac{5}{2}} + \frac{21}{2}x^2 - 16x$ .                                                                                                                                                                                                                                                 |
|          | Use of limits 1 and 16                                                                                                                                          | <b>DM1</b>   | Limits either way round. Minimum $-7.5 - 384$ .<br>E.g. $\left(\left(\frac{1}{2} - 16\right) - (2 - 10)\right) - ((128 - 256) - (2048 - 2560))$<br>$= (15.5 + 8) - (-128 + 512)$<br>$= -7.5 - 384 = -391.5$<br>Or<br>$(-2 + 10.5 - 16) - (-2048 + 2688 - 256)$<br>$= -7.5 - 384 = -391.5$<br>Or<br>$-504 + \frac{225}{2} = -391.5$ |
|          | $391.5$ or $\frac{783}{2}$                                                                                                                                      | <b>A1</b>    | CAO<br><b>SC B1</b> for correct answer if M1 B1 B1 DM0 scored.                                                                                                                                                                                                                                                                     |

| Question | Answer                                                                                                  | Marks      | Guidance                                                       |
|----------|---------------------------------------------------------------------------------------------------------|------------|----------------------------------------------------------------|
|          | <b>Alternative Method for Question 4</b>                                                                |            |                                                                |
|          | Height of triangle = 15                                                                                 | <b>B1</b>  |                                                                |
|          | Area of triangle = $\frac{1}{2} \times 15 \times \text{their height} = 112.5$                           | <b>M1</b>  |                                                                |
|          | Integrates $\pm \left( 5x^{\frac{3}{2}} - 20x \right)$ to $\pm \left( 2x^{\frac{5}{2}} - 10x^2 \right)$ | <b>B1</b>  |                                                                |
|          | Use of limits 1 and 16 on their integral and subtracts area of triangle                                 | <b>DM1</b> | Limits either way round.                                       |
|          | 391.5                                                                                                   | <b>A1</b>  | CAO<br><b>SC B1</b> for correct answer if M1 B1 B1 DM0 scored. |
|          |                                                                                                         | <b>5</b>   |                                                                |

| Question | Answer                                                                  | Marks         | Guidance                                                                                                                                   |
|----------|-------------------------------------------------------------------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| 5(a)(i)  | $32 - 80px + 80p^2x^2$                                                  | <b>B2,1,0</b> | B2 for all correct.<br>B1 for any two correct.<br>May be in a list. Ignore terms with higher powers.                                       |
|          |                                                                         | <b>2</b>      |                                                                                                                                            |
| 5(a)(ii) | $1 - 2x + \frac{3}{2}x^2$                                               | <b>B2,1,0</b> | OE<br>B2 for all correct.<br>B1 for any two correct.<br>May be in a list. Ignore terms with higher powers.                                 |
|          |                                                                         | <b>2</b>      |                                                                                                                                            |
| 5(b)     | Coefficient of $x^2 = 32 \times \frac{3}{2} + (-2 \times -80p) + 80p^2$ | <b>*M1</b>    | 3 terms FT <i>their</i> values.                                                                                                            |
|          | $48 + 160p + 80p^2 = 93 \Rightarrow 5(4p + 9)(4p - 1)[= 0]$             | <b>DM1</b>    | Set <i>their</i> 3-term coefficient to 93 and attempt to solve by factorising or other accepted method for solving their 3-term quadratic. |
|          | $p = -\frac{9}{4}$ and $p = \frac{1}{4}$                                | <b>A1</b>     | <b>SC B1</b> following M0 if method for solving quadratic is not shown.                                                                    |
|          |                                                                         | <b>3</b>      |                                                                                                                                            |

| Question | Answer                                                                                        | Marks        | Guidance                                                                                                                                                     |
|----------|-----------------------------------------------------------------------------------------------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6(a)     | $2x^2 - 2x(11x + 3) + 2 [= 0]$                                                                | <b>*M1</b>   | Substitutes $k = 2$ , $p = 11$ and eliminates $y$ or $x$ .<br>Note: $\frac{2x^2 + 2}{2x} = 11x + 3$ .                                                        |
|          | $-20x^2 - 6x + 2 [= 0] \Rightarrow [(5x - 1)(2x + 1) = 0]$                                    | <b>DM1</b>   | Simplifies to a 3-term quadratic.<br>Terms need not all be on one side.                                                                                      |
|          | Coordinates $\left(-\frac{1}{2}, -\frac{5}{2}\right), \left(\frac{1}{5}, \frac{26}{5}\right)$ | <b>A1 A1</b> | A1 for either both $x$ -values correct or for both coordinates of one point correct.<br>Need not be written as coordinates.<br>Fractions must be simplified. |
|          |                                                                                               | <b>4</b>     |                                                                                                                                                              |



| Question | Answer                                                              | Marks      | Guidance                                                                                                                                                                      |
|----------|---------------------------------------------------------------------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6(b)     | $2x^2 - kx(4x + 3) + 2 = 0 \Rightarrow (2 - 4k)x^2 - 3kx + 2 [= 0]$ | <b>*M1</b> | Substitute and reduce to 3-term quadratic.<br>Terms need not all be on one side.<br>Allow $2x^2 - 4kx^2 - 3kx + 2 [= 0]$ .                                                    |
|          | $(2 - 4k)x^2 - 3kx + 2 [= 0]$                                       | <b>A1</b>  | Correct quadratic.<br>All terms to one side.<br>Allow $2x^2 - 4kx^2 - 3kx + 2 [= 0]$ .                                                                                        |
|          | $b^2 - 4ac = 9k^2 - 4 \times (2 - 4k) \times 2$                     | <b>DM1</b> | Use of $b^2 - 4ac$ . Must be correct for <i>their</i> $a$ , $b$ , $c$ .<br>$a$ term must have two components.                                                                 |
|          | $9k^2 + 32k - 16 [< 0] \Rightarrow (k + 4)(9k - 4) [< 0]$           | <b>M1</b>  | Attempt to solve a 3-term quadratic in $k$ by factorising or other accepted method for solving their 3-term quadratic.                                                        |
|          | $-4 < k < \frac{4}{9}$                                              | <b>A1</b>  | <b>SC B1</b> following M0 if no method shown for solving quadratic.<br>A0 for correct answer following incorrect quadratic.<br>Must be $k$ .<br>Allow other correct notation. |
|          |                                                                     | <b>5</b>   |                                                                                                                                                                               |

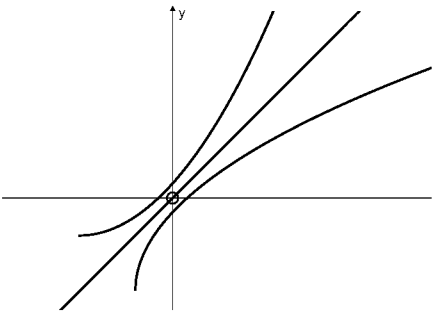
| Question | Answer                                                                                                  | Marks     | Guidance                                                                                                                     |
|----------|---------------------------------------------------------------------------------------------------------|-----------|------------------------------------------------------------------------------------------------------------------------------|
| 7(a)     | $\left[ \frac{dy}{dx} = \right] 8x - \frac{18}{x^3}$                                                    | <b>B1</b> | OE<br>Accept unsimplified.                                                                                                   |
|          | At $x = 2$ , $\frac{dy}{dx} = \frac{55}{4}$                                                             | <b>M1</b> | OE<br>For evaluating <i>their</i> $\frac{dy}{dx}$ or $\frac{dx}{dy}$ .                                                       |
|          | $\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx} \Rightarrow \frac{55}{4} = -5 \times \frac{dt}{dx}$ | <b>M1</b> | For correct use of chain rule with $\pm 5$ and <i>their</i> $\frac{dy}{dx}$ (may be algebraic).<br>Condone missing brackets. |
|          | $\left[ \frac{dx}{dt} = \right] -\frac{4}{11}$                                                          | <b>A1</b> | Or decreasing at a rate of $\frac{4}{11}$ .<br>AWRT $-0.364$ .                                                               |
|          |                                                                                                         | <b>4</b>  |                                                                                                                              |

| Question | Answer                                                                 | Marks     | Guidance                                                  |
|----------|------------------------------------------------------------------------|-----------|-----------------------------------------------------------|
| 7(b)     | $8x - \frac{18}{x^3} = 0 \left[ \Rightarrow x^4 = \frac{9}{4} \right]$ | <b>M1</b> | Equating <i>their</i> 2-term $\frac{dy}{dx}$ to zero.     |
|          | $x = \pm\sqrt{\frac{3}{2}}$ or $\pm\frac{\sqrt{6}}{2}$                 | <b>A1</b> | AWRT $\pm 1.22$ .                                         |
|          | $y = 4$ (for both)                                                     | <b>A1</b> | A0 A1 if one point correct.<br>AWRT 4.00.                 |
|          | $\left[ \frac{d^2y}{dx^2} \right] = 8 + \frac{54}{x^4}$                | <b>M1</b> | For differentiation.<br>At least one correct term needed. |
|          | So both are minima                                                     | <b>A1</b> | No need for reason.<br>WWW on $x$ -values.                |
|          |                                                                        | <b>5</b>  |                                                           |

| Question | Answer                                                                                                                                                                                                                                                                                          | Marks     | Guidance                                                                                                                                                                                            |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8        | Centre of circle is $(3, -5)$                                                                                                                                                                                                                                                                   | <b>B1</b> | Seen or implied.                                                                                                                                                                                    |
|          | $x = -2 \Rightarrow y^2 + 10y - 11 = 0 \quad [\Rightarrow (y-1)(y+11) = 0]$<br>or $(y+5)^2 = 36$ .                                                                                                                                                                                              | <b>M1</b> | 3-term quadratic.<br>Terms need not all be on one side.<br>$(3, -5)$ must be correct if it is used.                                                                                                 |
|          | $y = 1$ and $y = -11$                                                                                                                                                                                                                                                                           | <b>A1</b> | No method needed for solving quadratic.                                                                                                                                                             |
|          | Gradient of line $PC$ is $\frac{-5-1}{3-(-2)} \left[ = \frac{-6}{5} \right]$ or gradient of line $QC$ is $\frac{-5+11}{3-(-2)} \left[ = \frac{6}{5} \right]$                                                                                                                                    | <b>M1</b> | At least one correct.                                                                                                                                                                               |
|          | Gradient of tangent at $P$ is $\frac{5}{6}$ or gradient of tangent at $Q$ is $-\frac{5}{6}$                                                                                                                                                                                                     | <b>A1</b> | At least one correct.                                                                                                                                                                               |
|          | Equation of tangent is $y-1 = \frac{5}{6}(x+2)$ . Crosses $y = -5$ at $x = -\frac{46}{5}$ , so<br>distance = $\frac{36}{5}$<br>or equations of two tangents are $y-1 = \frac{5}{6}(x+2)$ and $y+11 = -\frac{5}{6}(x+2)$ and these<br>meet when $x = -\frac{46}{5}$ so distance = $\frac{36}{5}$ | <b>M1</b> | Distance from point of intersection to line<br>$x = -2$ is $6 \times \frac{6}{5} \left[ = \frac{36}{5} \right]$ .<br>Note $y = \frac{5}{6}x + \frac{8}{3}$ and $y = -\frac{5}{6}x - \frac{38}{3}$ . |
| 8        | Area = $\frac{1}{2} \times 12 \times \text{their } \frac{36}{5}$                                                                                                                                                                                                                                | <b>M1</b> | Condone use of $\frac{46}{5}$ .                                                                                                                                                                     |
|          | $\frac{432}{10}$ or $\frac{216}{5}$ or 43.2                                                                                                                                                                                                                                                     | <b>A1</b> |                                                                                                                                                                                                     |
|          |                                                                                                                                                                                                                                                                                                 | <b>8</b>  |                                                                                                                                                                                                     |

| Question | Answer                                                                                      | Marks      | Guidance                                                                                                                                                                      |
|----------|---------------------------------------------------------------------------------------------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9(a)     | $\frac{1}{2}r^2\alpha = 8\alpha \Rightarrow r = 4$                                          | <b>B1</b>  |                                                                                                                                                                               |
|          | $\frac{1}{2}r^2 \sin \alpha = 4 \Rightarrow \alpha = \frac{\pi}{6}$                         | <b>B1</b>  |                                                                                                                                                                               |
|          | Area of segment = $\frac{1}{2} \times 4^2 \times \frac{\pi}{6} - 4$                         | <b>M1</b>  | Using <i>their</i> $r$ and <i>their</i> $\alpha$ .<br>Condone $4 - \frac{1}{2} \times 4^2 \times \frac{\pi}{6}$ .<br>Allow use of $8 \times \text{their } \alpha - 4$         |
|          | $= \frac{4}{3}\pi - 4$                                                                      | <b>A1</b>  | Fraction must be simplified.<br>Allow $\frac{4\pi - 12}{3}$ .                                                                                                                 |
|          |                                                                                             | <b>4</b>   |                                                                                                                                                                               |
| 9(b)     | $r^2 + r^2 - 2r^2 \cos \alpha = \frac{1}{2}r^2 \Rightarrow 2 \cos \alpha = \frac{3}{2}$     | <b>*M1</b> | Or $\alpha = 2 \sin^{-1} \left( \frac{\sqrt{2}}{4} \right)$ or $2 \sin^{-1} \left( \frac{1}{2\sqrt{2}} \right)$<br><br>Using $\alpha = \frac{\pi}{6}$ or $r = 4$ implies 0/4. |
|          | $\alpha = 0.723 [0.72273 \dots] \Rightarrow \frac{1}{2}r^2 \sin(\text{their } 0.723) [= 4]$ | <b>DM1</b> | Or $\alpha = 41.4 \Rightarrow \frac{1}{2}r^2 \sin(\text{their } 41.4) [= 4]$                                                                                                  |
|          | $r = 3.48 [3.4777 \dots]$                                                                   | <b>A1</b>  | Accept $r^2 = 12.1$ AWRT.                                                                                                                                                     |
|          | Area of segment = $\frac{1}{2} \times 3.48^2 \times 0.723 - 4 = 0.371 [0.37068 \dots]$      | <b>A1</b>  | Or $\frac{1}{2} \times 3.48^2 \times \frac{\pi}{180} \times 41.4 - 4 = 0.371$ AWRT.                                                                                           |
|          |                                                                                             | <b>4</b>   |                                                                                                                                                                               |

| Question | Answer                                                                    | Marks         | Guidance                                                                                                                                                              |
|----------|---------------------------------------------------------------------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10(a)    | {Stretch} {factor 3} {parallel to $y$ -axis/in $y$ -direction/vertically} | <b>B2,1,0</b> |                                                                                                                                                                       |
|          | Translation $\begin{pmatrix} -2 \\ -5 \end{pmatrix}$                      | <b>B2,1,0</b> |                                                                                                                                                                       |
|          |                                                                           | <b>B1</b>     | Transformations correct and in the correct order.                                                                                                                     |
|          | <b>Alternative solution for Question 10(a)</b>                            |               |                                                                                                                                                                       |
|          | Translation $\begin{pmatrix} -2 \\ 5 \\ -\frac{5}{3} \end{pmatrix}$       | <b>B2,1,0</b> |                                                                                                                                                                       |
|          | {Stretch} {factor 3} {parallel to $y$ -axis/in $y$ -direction/vertically} | <b>B2,1,0</b> |                                                                                                                                                                       |
|          |                                                                           | <b>B1</b>     | Transformations correct and in the correct order.                                                                                                                     |
|          |                                                                           | <b>5</b>      | The translation parallel to the $x$ -axis can be made anywhere in the sequence.<br>Note: If 3 or more transformations are given then maximum 2/5 for any correct one. |

| Question | Answer                                                                            | Marks        | Guidance                                                                                                                                                                                     |
|----------|-----------------------------------------------------------------------------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10(b)    |  | <b>B1</b>    | For line or curve in correct quadrants only.                                                                                                                                                 |
|          |                                                                                   | <b>B1</b>    | Must not pass through (0, 0).<br>Fully correct including the line $y = x$ . No label needed.<br>Approximate reflection of $y = f(x)$ .<br>Curve must not come back on itself.                |
|          |                                                                                   | <b>2</b>     |                                                                                                                                                                                              |
| 10(c)    | $y = 3\sqrt{x+2} - 5 \Rightarrow \frac{y+5}{3} = \sqrt{x+2}$                      | <b>M1</b>    | Allow $x/y$ swap.                                                                                                                                                                            |
|          | $[g^{-1}(x)] = \left(\frac{x+5}{3}\right)^2 - 2$                                  | <b>A1</b>    | Must be in terms of $x$ .<br>Not ' $x = \dots$ '.                                                                                                                                            |
|          |                                                                                   | <b>2</b>     |                                                                                                                                                                                              |
| 10(d)    | [Range of $g^{-1}$ is $g^{-1}(x) \geq -2$ ]                                       | <b>B1 FT</b> | Following <i>their</i> $g^{-1}(x) = \left(\frac{x+a}{b}\right)^2 - c$ where $a$ , $b$ , $c$ are non-zero.<br>Not $x \geq -2$ . Not $> -2$ .<br>Accept other notations, e.g. $[-2, \infty]$ . |
|          |                                                                                   | <b>1</b>     |                                                                                                                                                                                              |

| Question | Answer                                                                                                                                             | Marks     | Guidance                                                                   |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----------------------------------------------------------------------------|
| 10(e)    | $[g^{-1}h(4) = g^{-1}(2)] = \frac{31}{9}$                                                                                                          | <b>B1</b> | AWRT 3.44.                                                                 |
|          |                                                                                                                                                    | <b>1</b>  |                                                                            |
| 10(f)    | $hg^{-1}$ is impossible since the <b>range</b> of $g^{-1}$ is $x \geq -2$ is <b>not within</b> the <b>domain</b> of <b>h</b> , which is $x \geq 0$ | <b>B1</b> | Minimum acceptable: ‘The range of $g^{-1}$ is not within the domain of h’. |
|          |                                                                                                                                                    | <b>1</b>  |                                                                            |